B2.2 Organelles and compartmentalization

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b2-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 244–247 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

B2.2.1 Organelles are discrete subunits with particular jobs

pp. 244–245

An **organelle** is a separate part of a cell, adapted to carry out one or more functions. Nuclei, vesicles, ribosomes and the plasma membrane all count.

Membranes	Examples
Two	Nucleus, mitochondrion, chloroplast
One	Vesicle, vacuole, lysosome, ER, Golgi apparatus
None	Ribosome

Watch out: the cell wall, cytoskeleton and cytoplasm are not organelles.

B2.2.2 Why the nucleus is kept separate from the cytoplasm

pp. 246–247

In eukaryotes, transcription happens in the nucleus and translation in the cytoplasm. mRNA can therefore be modified before it meets ribosomes. Prokaryotes have no nuclear membrane, and so ribosomes may start on mRNA as soon as it forms.

B2.2.3 Advantages of compartments in the cytoplasm

p. 247

Compartments **concentrate** enzymes and substrates, and **separate** processes that would interfere with one another. They also allow each region to keep its own conditions.

Lysosomes hold digestive enzymes behind a membrane at a low pH. The enzymes work only at this low pH, so any that leak into the neutral cytoplasm do little harm. **Phagocytic vacuoles** enclose engulfed food or pathogens, so that digestion takes place apart from the cytoplasm.

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